Dataset: A video dataset of a wooden box assembly process

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ABSTRACT

This paper presents a video dataset of a 9-step wooden box assembly process including 17 subjects. The main strength of this dataset is the design of a standard and uniform workflow and the use of multiple cameras capturing videos from two different viewpoints. 62 video files were collected with the total size of 20 GB and the total duration of 13 hours. Each of the video is complemented with temporal annotations that indicate the starting and ending timestamps of each work step in the assembly process. We also provide statistical descriptive analyses of the recorded processes. Our dataset can be utilized for developing solutions to human activity recognition, process documentation, and many others that involve human-object interaction.

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BACKGROUND & SUMMARY

Instructional videos are helpful when people need to learn new tasks ranging from common everyday activities such as cooking, assembling furniture to specific industrial applications such as machinery maintenance. In the past decades, the studies on analyzing and extracting useful information from instructional videos have been extensively investigated and spread a wide range of applications. Automatically recognizing human actions from massive instructional videos [1, 2, 8, 9] can help produce automatic text description to instruct people or machines how to achieve new tasks. Applying object detection methods to videos can enhance understanding of hand manipulation and hand-object interactions [3, 4] which is essential for fine-grained workflow description. In addition, learning from videos of human activities is used in related applications such as biometrics and human kinematics analysis [7]. In order to facilitate the research work in understanding of instructional videos, a few datasets have been collected and used. For example, datasets with videos of different cooking recipes have been collected [5, 11]. Alayrac et al. [1] collected a video dataset of five tasks including making a coffee, changing car tire, performing cardiopulmonary resuscitation, jumping a car and repotting a plant. Zhukov et al. [12] collected the CrossTask dataset with videos of 18 primary tasks including cooking, car maintenance, crafting, and home repairs as well as videos containing actions related to these tasks, for example, *pouring milk* and *pouring water* share the action but have a different utilized object. HowTo100m is a large-scale dataset of narrated video from Youtube, it is used to learn a dense video-text embedding [6].

However, existing datasets for analyzing instructional videos have some limitations. First, most datasets are collected from online video websites, such as YouTube, where users could upload and share their videos freely. The videos of the same activity might have steps conducted in different orders and do not follow standard workflow. Second, there is a lack of datasets providing videos taken by cameras from multiple positions. Videos captured from different viewpoints could provide complementary information for kinematics analysis and may enhance the performance of recognition algorithms.

In this paper, we present a new annotated video dataset that covers a process of wooden box assembly. We define nine steps for wooden box assembly workflow and invite participants to perform the assembly workflow. We use two cameras to capture the assembly processes from different viewpoints (illustrated in Fig.2a and 2b). We also label the temporal information of the nine steps in each video: timestamps of the beginning and end of every step. The dataset contains 62 videos of 17 subjects with a total duration of 13.0 hours. Such a dataset could benefit research work in different fields such as image processing, computer vision and robotics.

METHODS

Participants

We invited 17 participants (4 females and 13 males) for our data collection. Before conducting the process, each participant was introduced to the tools and materials used in the experiment and was informed of the potential risks. The participants gave their informed consent to participate in the study and have their videos published.

Data Collection Setup

The experiments were carried out in a studio for woodwork. The environment of the studio is shown as Figure 1. The tools used in the assembly included tape measure, pencil, drill, sandpaper, rubber hammer, tape, screwdriver, brush and tissue paper. The materials for making a wooden box included box edges, screws, hinges and a bottle of varnish. Tools and materials used in the experiments are labeled in Figure 1.

We used two GoPro Hero 7 Black action cameras to record the assembly process from different viewpoints. One camera was placed in front of the participants and the other was positioned on the

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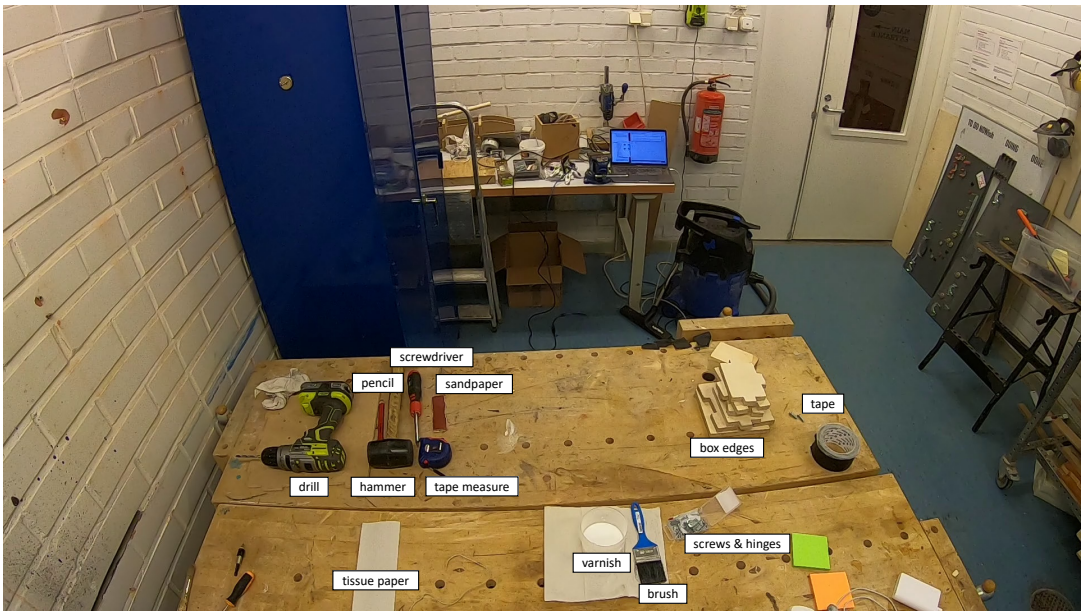

Figure 1: Environment of the Studio and tools and materials used in the experiments

upper right of the participants. Cameras were tilted downward to ensure that the operations done by the participants were fully captured. The viewpoints of the two cameras are shown in Figure 2a and Figure 2b. The data was acquired at 60 frames per second with 1920 * 1080 pixels resolution. For each assembly process performed by the participants, we recorded the data by the two cameras simultaneously.

(a) Cam 1

(b) Cam 2

Figure 2: Views of the two GoPro cameras

Procedure

The process of wooden box assembly consisted of nine steps. The first seven steps are the assembling procedures, and the last two - the disassembling:

- Measuring. The first step is to randomly select a position on the board for drilling. In practice, the participant used a tape measure to measure the distances (10-20cm) from the selected position to the edges of the board, and marked the selected point with a pencil. This step is the preparation step for drilling.
- Drilling. The second step is to drill a hole at the point marked in the first step. This step is for testing the drill.
- Sanding. The third step is to sand each of the box edges with the sandpaper.
- Assembling. The fourth step is to assemble the box edges with the rubber hammer.
- Screwing. The fifth step is to screw the top to the box edges with the screwdriver.
- Varnishing. The sixth step is to varnish the top using the brush. This step is to make the box surface smoother.
- Drying. The seventh step is to dry the varnish using tissue paper.
- Unscrewing. The eighth step is to unscrew the top to the box edges with the screwdriver.
- Disassembling. The ninth step is to disassemble the box into 5 original pieces with the rubber hammer.

We assign an activity id to each of the nine activities following the order in which the steps were performed . The ids, corresponding activities and tools are listed in Table 1.

Before the start of the experiment, the participants watched a demonstration video of the assembly process and were instructed

Table 1: The predefined wooden box assembly process

ID	Activity	Tools
1	Measuring	tape measure, pencil
2	Drilling	drill
3	Sanding	sandpaper
4	Assembling	rubber hammer
5	Screwing	screwdriver
6	Varnishing	brush
7	Drying	tissue paper
8	Unscrewing	screwdriver
9	Disassembling	rubber hammer

about the proper operations of each step. After they fully understood the steps of wooden box assembly, they were asked to perform the whole process in one take. An instructor was there to guide them through the process, and participants were allowed to stop, if necessary.

Data Processing

After data collection, we performed data processing on the videos. The cameras were connected to the Macbook Pro laptop (macOS High Sierra Version 10.13.6 operating system, Intel Core i5 processor), which was used for both processing and annotation of the videos.

The main purpose of data processing was to edit interrupted videos. There were two types of camera interruptions during the shooting sessions. The first one was that the cameras truncated the video automatically and stored video clips if the video was too long. In this situation, we directly added the second clip to the end of the first one to merge them into one complete video. The second one happened when the camera was interrupted due to malfunctions such as battery depletion. We continued to shoot the unfinished steps after the equipment malfunctions were resolved. In this situation, we merged the two interrupted clips and cut the video taken by the other camera into the same time length as the previous merged video.

We changed the video resolution to 1280×720 and blurred all the faces in the videos using a deep learning-based face detector and OpenCV blurring function.

Data Annotation Process

After processing the videos, we marked the temporal extent of the nine activities in each filmed process. Annotations were performed by one annotator observing the videos manually. The starting time of an activity was identified as the moment of contact of the subject's hands with an object (a tool or the material). The finishing time was recorded when the subject released an object. The annotations were reviewed three times: twice by the same annotator and one more time by another annotator to guarantee the correctness. Detailed format of the annotation is given in Table 2.

DATA RECORDS

Data records introduced in this paper along with the annotation files are available through the Zenodo repository [10]. As each subject is assigned a unique integer ID ranging from 01 to 17, the videos and annotation files are organized by subject ID and arranged into 17 directories. The details of the data are illustrated below.

Video Data

In each directory, there are two pairs of videos in MP4 format (1280x720 resolution), representing the process performed with the sensor glove and without. The sensor glove visually differed from a normal protection glove as it had special sensors and microelectronics attached. Each pair of the two videos were taken by the two cameras for the same assembly process simultaneously. The four videos are named as:
n_glove_cam1.mp4
n_glove_cam2.mp4
n_noglove_cam1.mp4
n_noglove_cam2.mp4.
In the names of the files, *n* represents the subject ID number, "glove" represents that the assembly process was performed with the sensor glove on, while "noglove" represents the process was performed without the sensor glove. "cam1" represents that the video was recorded by the camera positioned in front of the subjects, and "cam2" represents the video was recorded by the camera placed on the upper right of the participants.

To protect privacy of the participants, we removed the audio information of the videos and blurred the participants' faces.

Annotation Records

The annotation information of each filmed process is provided as a CSV file named as n_noglove.csv or n_glove.csv, where n is the ID of the subject who performed the process. n_noglove.csv annotates the process done by subject n without the sensor glove, and n_glove.csv annotates the process done by subject n with a glove.

Table 2 shows an example of annotation file (01_glove.csv). There are 3 columns in each annotation file. The first column is the activity ID. The second and third columns are the starting time and the finishing time of the activity. The time is recorded in seconds.

Table 2: An example of annotation file

1	23.12	40.25
2	44.39	57.5
3	69.21	124.40
4	126.05	169.57
5	172.10	312.24
6	318.10	332.00
7	339.10	356.00
8	361.03	430.21
9	431.40	459.34

Missing Data

A directory with complete data should contain four videos and two annotation files. However, there are instances of missing video data. Due to equipment malfunctions, some shooting sessions were interrupted. These uncompleted videos were not included in the dataset. As a result, in two of the directories (subject ID 02 and 11), there are only three videos, missing the video taken by one camera in one of the assembly process. In addition, because of the time limitation of two participants (subject ID 05 and 07), they didn't finish both of the two assembly processes (with and without the sensor glove). These two subjects have two videos for only one assembly process taken by both of the cameras and one corresponding annotation file. The numbers of videos in different setting in our dataset are listed in Table 3.

Table 3: Number of videos in different settings

Camera no.	W/o the sensor glove	With the sensor glove on
1	15	15
2	16	16

TECHNICAL VALIDATION

In this section, we provide statistical evaluation of our dataset. We first analyze the duration of the whole assembly process and workflow steps. Then we validate the synchronization of the videos taken by the two cameras during the same processes.

Duration of the workflow

The total length of the videos in our dataset is 13.0 hours with an average duration of 12 minutes and 45.72 seconds per video. We calculate the statistics of duration of the whole process in different settings. As shown in Table 4, the duration of the whole processes finished without the sensor glove and with the sensor glove on is 781.85 ± 365.74 seconds and 749.59 ± 308.47 seconds (containing 95% data) respectively. Since every participant was asked to perform the workflow once without the sensor glove first, and once again with the sensor glove on, they are more proficient during the second session than during the first session, so the duration of the workflow finished with the sensor glove on is shorter than without.

Table 4: Statistics on duration of the whole assembly session (sec.)

Setting	Mean	St. Dev.	Min	Median	Max
W/o glove	781.85	232.76	473.47	754.99	1314.88
With glove	749.59	216.57	468.71	713.67	1252.61
Total	765.72	225.39	468.71	744.09	1314.88

Then, we analyze the duration of each step of the workflow. Table 5 shows the results. The sum of duration of each workflow step takes up 90.2% of the total duration, leaving less than 10% for the random states (such as looking for the tools). We also compare the duration of each step.

The top 5 steps with the longest duration are screwing, unscrewing, assembling, drilling and sanding. Some subjects tend to spend longer time on each step while others tend to spend less time on each step.

Table 5: Statistics on duration of each workflow step

Workflow	Mean	St. Dev.	Min	Median	Max
Measuring	24.66	7.67	9.42	23.37	47.42
Drilling	68.9	66.89	15.83	49.78	294.31
Sanding	66.24	25.7	22.09	65.59	130.22
Assembling	75.72	28.28	32.45	66.57	143.96
Screwing	255.25	94.4	135.24	238.47	461.23
Varnishing	27.08	11.72	8.74	23.34	50.93
Drying	20.65	6.5	8.6	18.7	37.2
Unscrewing	105.45	27.87	50.8	101.11	177.57
Disassembling	47.11	20.47	23.67	41.17	104.16

Synchronization of the Cameras

During the filming sessions, the synchronization of the cameras was controlled manually by pressing the buttons of the two cameras to start and end shooting at the same time. To validate the synchronization of the cameras, we checked the durations of the two videos for each assembly process taken by the two cameras. The time difference between the two videos for one same assembly process is 0.32 ± 0.42 s. The time difference is less than one second which is negligible.

USAGE NOTES

The data can be used for developing and evaluating machine vision algorithms The dataset provides a coherent process of assembly with a strict order of steps. The main features of this dataset are realistic industrial set and background, variety of tools used in the process and multi-camera perspective. Zenodo repository includes a script *use_example.ipynb* that loads a video sample and annotations.

Records of 17 different subjects allow perform the cross-user evaluation of the workflow extraction algorithms. Indeed, the technical validation demonstrated that users performed the tasks in a different manner: some completed the task in 8 minutes, whereas some spent 20 minutes.

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AUTHOR CONTRIBUTIONS

Jiayun Zhang recruited subjects, acquired data, performed data labeling, processing and analysis, and wrote the paper during the visit to Aalto University. Petr Byvshev conceived and planned the project, supervised the data collection, processed and reviewed the data and wrote the paper. Yu Xiao conceived and planned the project and reviewed the paper.

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